Exponential Search

2	4	5	6	11	13	14	15	56	70
0	1	2	3	4	5	6	7	8	9

or if we find $\boxed{x = 13}$ → search space would be this.

intuition → reduce the search space then apply BS on it.

step

```
if (a[0] == x) return 0;
int i = 1;
while ( i < n && arr[i] ≤ x)
{
    i = i * 2;
}
return BS(a, i/2, min(i, n-1), x);
```

$\boxed{n = 10}$

3 ≤ 13
5 ≤ 13
11 ≤ 13
13 ≤ 13
15 ≤ 13 → terminate

8/2

2	3	5	6	11	13	14	15	56	70
0	1	2	3	4	5	6	7	8	9
	i	i							

$\boxed{l = 16}$

i=1

i=10 56 ≤ 56
while (i < n && arr[i] ≤ x)
i = i*2;

why min(i, n-1) → if x = 56, then l = 16

So, BS (a, i/2, 16, n)
8 (start), 16 (end)

but 16 is invalid index, so in that case we should pass n-1 index.

BS ⇒
[11, 13, 14, 15, 56]

use ⇒ i*2 ⇒ i<<2;
i/2 ⇒ i>>2;

⇒ left shift "<<" & Right shift ">>"

ex ⇒ 21 ⇒ Binary
111 → 7
1111 → 15
$8 + 2^2 + 2 + 1$
$8 + 4 + 2 + 1 = 15$

1111
$16 + 15 = 31$ ⇒ 10101

21
16
0 1

⇒ a = 21 ⇒ 10101
a = a << 1 (left shift)
= a = a * (2^1) = 21 * 2 = 42

MSB LSB
1 0 1 0 1 0 → add 0 at last because of shifting

Now,
4 3 2 1 0
1 0 1 0 1 0 → 0 1 0 1 0
⇒ 32 + 8 + 4 = 42

10 ⇒ if a size only 5 bits (where 5th index 1 will discarded)

Right shift ⇒ 21 ⇒ 1 0 1 0 1
a = a / (2^1) = 21/2 = 10.5 = 15

⇒ 1 0 1 0 1
0 1 0 1 0
⇒ 0 1 0 1 0 1 = 0 1 0 1 0
4 3 2 1 0

discarded

$2^3 + 2$
= 10 Ans

So, Points to remember ⇒

① for left shift (*)
↳ if size is fixed as size of a (input)
then in that case MSB will be discarded & add 0 at LSB.
↳ else consider whole MSB to LSB.

② for Right Shift (/)
↳ Surely, discard the LSB and add 0 at MSB.

⇒ T.C ⇒

1	2	...	4	...	8	16	...	32
2^0	2^1		2^2		2^3	2^4		2^5

$m = 2^n$ form

① ② ③ ④ → 4 steps

log 16 = log (i) = $\log_2 2^4$ = 4 steps

⇒ log(m) → to find subarray (i needed)

⇒ BS on subarray ⇒ $2^4 - 2^3$ ⇒ BS(a, i/2, i, x)
16 − 8 = 8 total elt in subarray

⇒ $2^{\log(m)} - 2^{\log(m-1)}$

why $2^{\log(m)}$ ⇒ as we same above,
→ $2^{\log(m)}$ m = 16
$2^{\log 16}$ = $2^{\log 2^4}$ = 2^4

$$\Rightarrow 2^{\log(m)} - 2^{\log(m-1)}$$
$$= 2^{\log(m)} - 2^{\log(m)} \cdot 2^{-1}$$
$$= 2^{\log(m)} \cdot (1 - 2^{-1})$$
$$= O\left(\frac{2}{2}^{\log(m)}\right)$$